Part I: create a list of basal attributes

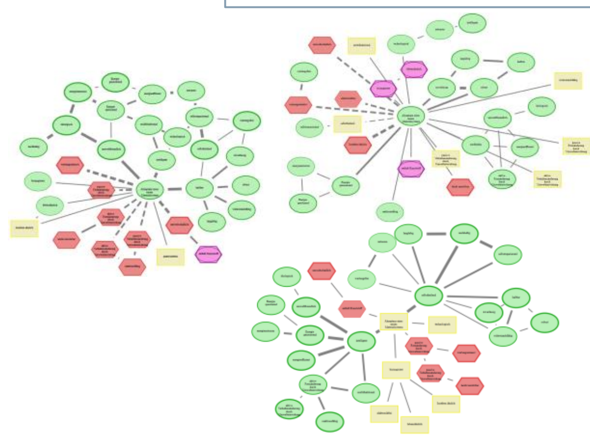
(a) Conducted 14 guided expert interviews to create an initial list of basal attributes

(b) Reduced initial lists by three raters

(c) Got ratings of 21 experts for (a) the relevance of remaining basal attributes and (b) possible missing's

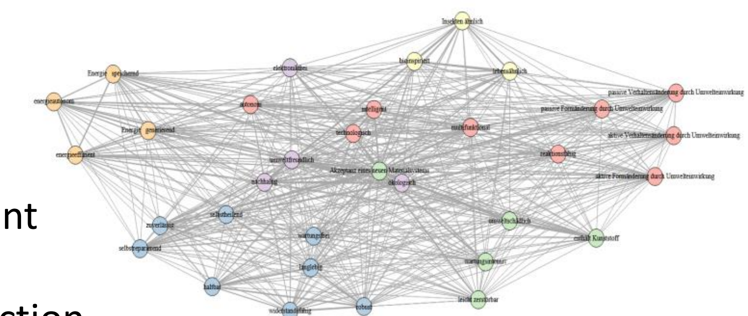
(d) Automatically scanned all *liv*MatS for possible relevant adjectives using part-of-speech tagging

Part II: Cognitive-Affective Maps study



Conducted a large-scale CAM study (N=192) to assess lay perceptions.

Applied analyses on different levels of the CAMs, using especially community detection algorithms



Identified clusters of basal attributes surrounding the central concept

Part III: proof of concepts - textual descriptions of identified clusters

LLM-assisted generation of expert-refined scenarios for future material systems using attribute clusters.

Insect-like Building Facade

Bio-Smart is a bio-inspired building facade.

The surface of the new Bio-Smart facade is easily damaged and therefore requires frequent maintenance. Despite its environmentally harmful components, since it contains plastic, Bio-Smart impresses with its life-like properties.

The innovation lies in transferring life-like characteristics from insect surfaces to the surface of the building facade.

Scenario 1 (Cluster 3+6)

Conducted large-scale study (N=298) assessing lay perceptions of six material system descriptions across usefulness, trustworthiness, usage intention, and emotional response.